QR-007 Work orders: steps, buy-offs, holds and issues

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Title	Work orders: steps, buy-offs, holds and issues
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Derived from CS-013, CS-008 and CS-007. If this and the parent disagree, the parent is right. App labels are copied from `app/workorders.js` and `app/reports.js`.

1. Part or run: what is the difference, and which do I make?

A part (`P-SN6-###`) is the thing the car needs. A run (`WO-SN6-###`) is one attempt at making it; the Work Orders tab calls a run a work order. Make the part first, then press + New run on it (CS-013 §4; §7.1).

Also called: WO, work order, run, traveler, traveller, job; part, P number; remake, re-lay, second attempt.

A part has many runs, and each run belongs to one part (CS-013 §4). A re-laid part is a new run on the same part, never a new part: dispose the issue on the failed run as Scrap, then press + New run (CS-013 §4.1 rule 3). The part keeps its `P` ID and every label stays valid.

You are making	You should make (CS-013 §2)
A part for the car	A part on the Parts tab, then a run
Something with a mold, a checklist or a cure to sign, not for the car	A part and a run started from the R&D tab (an R&D part)
Coupons, offcuts or trial panels you will cut, measure and compare	A study on the R&D tab, with no work order
A one-off job that will never be remade	New WO on the Work Orders tab (a run with no part)

An R&D part keeps every blocker, hold and evidence check: they work exactly as on a season part. It is kept off the Season tab and the Parts and Work Orders lists and lives on the R&D tab (CS-013 §4).

2. Starting a run

On the Parts tab press New Part (or open the part), fill its requirements, then press + New run. The run gets `WO-SN6-###` and copies the part's plan: engineers, deadline, mass target, technique, stack and BOM (CS-013 §7.1).

Also called: new work order, new WO, new run, create a work order, start a job.

1. Make the part on the Parts tab with New Part, or open the part if it exists.
2. Fill the part's requirements before any work: Subteam, Layup type, Layup deadline, Mold Engineer, Manufacturing Engineer, Target weight (g), the Layup stack (CS-002 §7.1) and Materials (plan).
3. Press + New run on the part.
4. For a remake, choose Use a previous run and pick what to keep. Start fresh copies the part's own plan.

5. On the run, set the quality checks with numeric targets in Quality (+ check). Mass is prefilled from Target weight.

The run's steps come from its technique, which the Layup type picks (CS-013 §7.1 step 5). If the checklist does not fit the job, fix the technique, not the run: pick the right Layup type before the run starts, or ask a lead to add a technique. Members cannot delete steps. A step that does not apply is signed with a note saying why.

Person fields (Mold Engineer, Manufacturing Engineer, Laid up by) are picked from the app's roster, not typed (CS-013 §4).

3. Who can sign a step?

Anyone signed in on the roster can buy off a step, including their own work, with two exceptions: a step that needs training needs someone a lead has recorded as trained, and Mold design review needs someone other than the mold's designer (CS-013 §4).

Also called: buy-off, buy off, sign-off, sign off, sign, initials, approve a step, who can sign.

The button on a step reads "buy off as [your name]". The app stores the signer's name, account, and the date and time. Treat it as "who was signed in when this was clicked", not a notarised signature (CS-013 §4).

1. Training to sign: the step can be bought off only by someone holding that training on the People tab. A lead may sign without it and must write why; the reason goes in the run's event log (CS-013 §4).
2. Independent design review: Mold design review shall be bought off by someone other than the part's Mold Engineer (CS-013 §4). The app only warns ("Self-review", with Sign it anyway) when the run's Mold Engineer signs; it never refuses, so the signer still checks who drew the CAD.
3. View as guest changes nothing, because every buy-off carries a name (CS-013 §5).

Each step gets a buy-off, a note and photos. Press the camera on the step row (Add photos to this step) to attach photos from your phone; a photo is never required (CS-013 §7.2 step 2).

Which training each step needs is in QR-007 §4.

4. Which training does each step need?

Mold design review needs Mold design. Glue and machining steps need ShopSabre CNC. Infusion steps need Resin infusion. Wet layup steps need Wet layup. Shape foam core needs Foam core (CS-013 §7.5 table).

Also called: training, trained, qualification, signed off to use, CNC training, ShopSabre training, infusion training.

Technique	Step	Training (CS-013 §7.5)
Mold infusion, Mold wet lay	Mold design review	Mold design
Mold infusion	Glue mold stock; Machine mold	ShopSabre CNC
Mold wet lay	Glue and machine mold	ShopSabre CNC
Mold infusion, Glass infusion	Dry stack and bag; Drop test; Infuse	Resin infusion
Mold wet lay	Wet layup and bag	Wet layup
Foam wrapped	Shape foam core	Foam core
Foam wrapped	Wet layup over core	Wet layup

A lead records who holds a training on the People tab. A lead is not trained by being a lead (CS-013 §4). The app's ShopSabre CNC training mirrors the RFS record, which CS-005 §4 names as the authority.

Every other built-in step (Stack frozen, Seal and release mold, Cure and demould, Trim and finish and the rest) needs no training. Their checks are blockers, evidence and the cure hold (QR-007 §5 and §6).

The table is the built-in techniques as the app shipped them on 2026-10-10. A lead can edit techniques; Reports → Layup techniques shows the live ones (CS-013 §7.5).

5. Why can't I sign this step? Blockers, evidence and OnHold

A blocker step blocks every step after it until someone buys it off. The blockers are Stack frozen, Mold design review, the drop test, and the acceptance criterion on an Other run. Some steps also need evidence before they can be signed (CS-013 §4).

Also called: blocked, blocker, locked step, greyed out, can't buy off, evidence, stack freeze, drop test, design review, OnHold, on hold, parked.

The run shows "Blocked by an unsigned blocker:" and names the step. Sign that one first.

Step	What it needs before it can be signed (CS-013 §4, §7.5)
Stack frozen	A layup stack on the run
Mold design review	The CAD, attached or linked, plus Mold design training
Machine mold, Glue and machine mold	A note
Drop test, 1 inHg or less over 10 min	A note with the reading (the number is CS-006 §7.4's)
Define acceptance criterion (Other)	A note: target and method, set before work starts

A lead can sign without the evidence, with a written reason (CS-013 §4).

A technique that lays up a stack shall keep Stack frozen ahead of the first physical step, and a drop-test blocker ahead of the resin step (CS-013 §7.5 step 4). Mold wet lay and Foam wrapped have no drop-test step; on those the start and end readings go in the note on "Wet layup and bag" ("Wet layup over core" on a Foam wrapped run) (CS-007 §7.5).

OnHold is a status a person sets to park a run. It blocks nothing in the app (CS-013 §4).

6. Why is the demould (demold) step locked? Still curing

Cure and demould (or Cure) stays locked until the FEB hold has run from the finish time on the step before: 48 h IN2 + AT30 SLOW, 24 h FAST, 24 h West 206 or 205, 36 h West 209 (CS-008 §5). Sign early and you get "Still curing" (CS-013 §4; CS-008 §5.1).

Also called: demould, demold, de-mould, cure hold, FEB hold, wait time, still curing, locked, how long to cure, on hold, step on hold.

Signing the step that starts a cure ("Infuse", "Wet layup and bag", "Wet layup over core") asks for "Resin mixed", "Finished on" and "at", and "Shop temperature, °C (optional)". The next step stays locked until the FEB hold for that resin has run (CS-008 §5.1).

The FEB hold is longer than the datasheet figure on purpose. It is a team decision, signed off by Simon Starbuck on 2026-08-01, not a manufacturer claim (CS-008 §5.1). XCR's hold, the datasheet figures and the reasons are in QR-001 and CS-008 §5.1. On the step, the "why N h?" link explains the number.

If the recorded shop temperature is below the datasheet's reference, the app warns "Colder than this number assumes, so it will run long. Test the flange, not the clock." It leaves the hold unchanged (CS-008 §5.1).

Only a lead can sign through a hold, from "Demould early?", which asks "Why are you demoulding early?". The reason goes in the event log with the lead's name (CS-013 §4).

The hold reads only the step directly before it. A drop-test step placed between "Wet layup and bag" and "Cure and demould" would switch the hold off, so on a wet layup the drop test goes in the "Wet layup and bag" note (CS-007 §7.5).

7. What to do when something goes wrong: raising and closing an issue

Raise it on the run it holds up: Raise an issue at the foot of the run's Issues section, or (Report an issue on this step) on the step where it happened. Close it with a Disposition, What happened and What was done (CS-013 §7.6).

Also called: issue, ticket, problem, defect, NCR, nonconformance, disposition, scrap, rework, UAI, use as is.

1. Raise the issue with Raise an issue, or on the step.
2. To close it, open the run's Issues section, or the issue's page.
3. Pick a Disposition: UAI (Use As Is), Corrective Action, Rework, Scrap or Other.
4. Fill What happened, the root cause.
5. Fill What was done. Scrap asks "Why could it not be saved?"; UAI asks "Why is it acceptable as it is?".
6. Press Resolve (Resolve issue on the issue's page).

A false alarm is cancelled with Cancel ticket (false alarm), not disposed (CS-013 §7.6 step 4).

If a fix did not hold, press Reopen on the issue's page. It withdraws the disposition and clears What was done, keeps What happened, and the run is held again until the issue is disposed a second time (step 5).

A run shall not complete while any of its issues lacks a Disposition, What happened or What was done (step 3). The app checks only Disposition and What was done, so the person closing out checks What happened.

8. Closing a run

Fill the actuals in Quality (mass, thickness, Ω table), check the photos, and set the status to Complete. If an issue is still open, the app opens the closeout list instead; dispose each one (CS-013 §7.3).

Also called: close a work order, complete, finish a run, closeout, done, mark complete.

1. Fill the actuals in Quality: mass, thickness, Ω table and the rest.
2. Check the photos.
3. Set the status to Complete.
4. If the closeout list opens, press Resolve on each issue, or Resolve all & complete work order at the foot (CS-013 §7.6 step 3).
5. If the part is safety or inspection relevant, print one copy for the physical binder at RFS (CS-013 §7.3 step 3).

There is nothing to export by hand. Every save writes straight to the shared database (CS-013 §7.3 step 2). The archive is a lead's monthly Backup into the team Drive (header button; on a phone, Backup database in the menu).

Which lots went in is asked at the infuse or wet layup buy-off: fabric roll, resin lot and hardener lot, pre-filled with whatever is open. "I don't know / not recorded" is a valid answer and is recorded as such. A confident wrong lot is worse than an honest gap (CS-013 §7.2 step 4).

9. I deleted something by mistake: Recently deleted

Go to Reports → Recently deleted and press Restore on the row. A deleted record stays there for 30 days, and restoring it brings back everything deleted with it, uploads included (CS-013 §7.7).

Also called: deleted, delete, undo delete, restore, bin, trash, recycle bin, recover, lost record; archive, archived.

1. Open the Reports tab.
2. Find the record under Recently deleted. Each row says who deleted it and how many days are left.

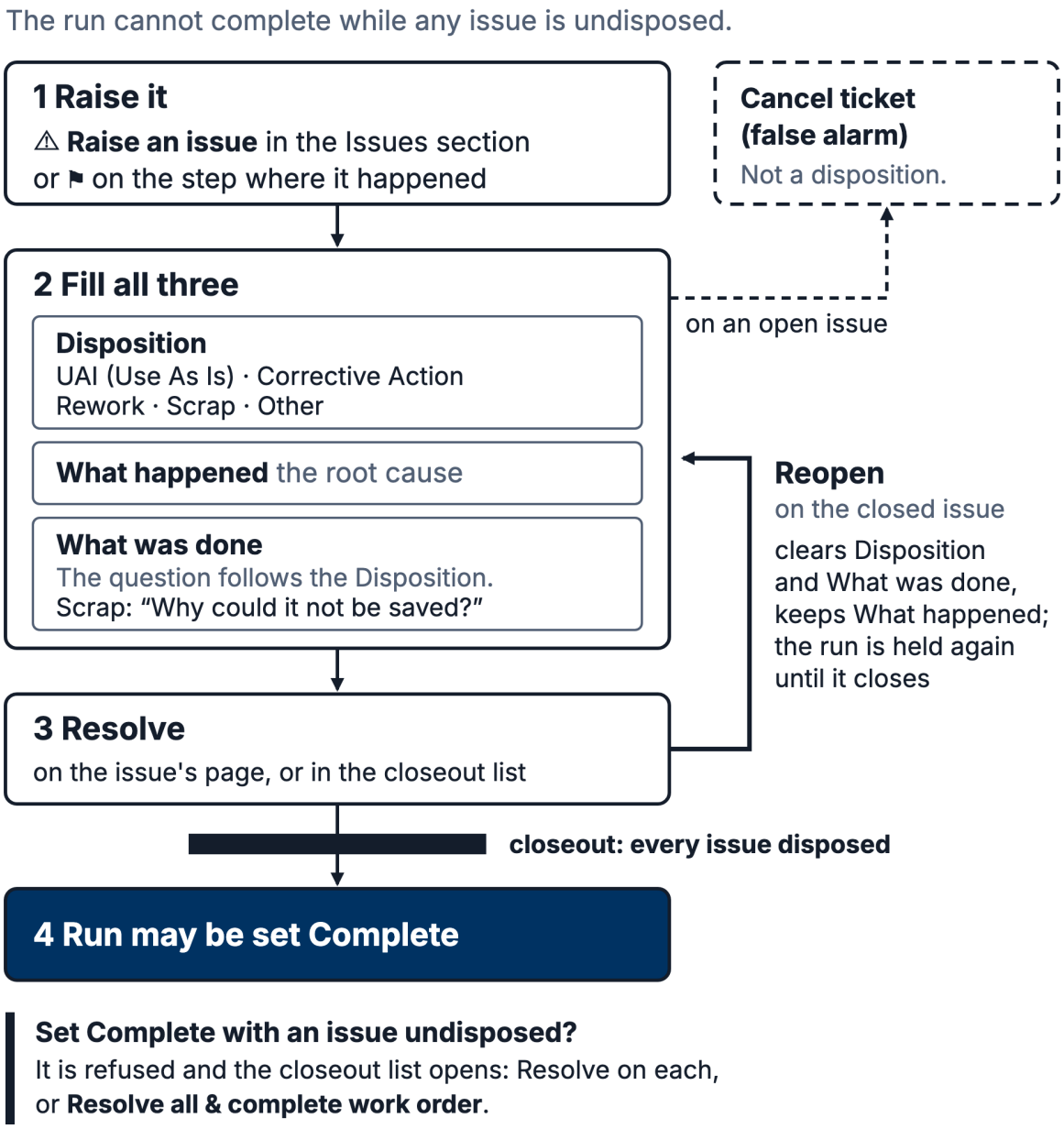


Figure 1: The issue path on a run: raise it (Raise an issue, or on a step), then Disposition, What happened and What was done; the app holds Complete until every issue has a Disposition and What was done. Reopen loops back and clears the disposition.

3. Press Restore.

Only a lead deletes parts, runs and issues. Deleting a run takes its issues and uploads with it (CS-013 §7.7 step 2). After 30 days a lead presses Empty them, and the records are gone for good (step 4). A deleted ID is never reissued (step 5).

A finished or abandoned part or run should be archived, not deleted (step 1). Any member can Archive one on its page in edit mode, or several with Select… on the list. An archived record leaves the lists, the Dashboard and the Season tab but keeps every field. The archived chip on the list shows them, each with Restore.

Delete is for a record that should never have existed (CS-013 §7.7 step 2).